Discovery report for Multitask ADMET Property Prediction For Drug Discovery

Research Objective

Develop an excellent ML model for predicting the following properties:

LogD: Measures compounds lipophilicity at a specific pH. Drugs typically fall into a well defined LogD range that balances aqueous solubility with membrane permeability, making understanding changes in LogD across a chemical series important to medicinal chemists. Additionally, assessing LogD of candidate molecules can suggest whether candidate molecules are "efficient" for their lipophilicity (lipophilicity generally has a linear relationship with affinity). Kinetic Solubility (KSOL): Measures how much a compound can be dissolved under non-equilibrium conditions. Helps screen compounds that will fail due to poor absorption or low bioavailability. Human Liver Microsomal (HLM) stability: Supports understanding of a compound's susceptibility to liver metabolism and can be used to predict in-vivo clearance of a candidate molecule. Measured using human liver microsomes and reported as intrinsic clearance for the compound Clint (mL/min/kg). Mouse Liver Microsomal (MLM) stability: Also used to predict in-vivo clearance of a candidate. The study of both MLM and RLM can provide a more comprehensive understanding of a compound's metabolic profile and how a compound might behave in multi-species preclinical development. Caco-2 Papp A>B: Measures the rate of flux of a compound across polarized Caco-2 cell monolayers from the apical (intestinal lumen-facing side) to basolateral (blood-facing side). This effectively mimics the absorption of a drug across the intestinal wall. Caco-2 Efflux Ratio: Measures the rate of flux of a compound across polarized Caco-2 cell monolayers. The Efflux Ratio is determined by a ratio of the apparent permeability coefficient (Papp) in both directions. Ratios of ~1 indicate that a compound primarily traverses the cell membrane via passive (diffusional) transport. Ratios > 2 generally indicate active transport of the compound across the cellular membrane by membrane bound transporters (e.g. efflux by p-glycoprotein). Mouse Plasma Protein Binding (MPPB): Determines the concentration of free drug in plasma (as % Unbound). Drugs that are not bound to plasma can bind to target proteins and yield the desired therapeutic effect, making this parameter crucial to understanding drug distribution. Mouse Brain Protein Binding (MBPB): This measures the fraction of a drug not bound to proteins within brain tissue. The unbound fraction of a drug in the brain is considered pharmacologically active and able to interact with central nervous system (CNS) targets. MBPB helps assess CNS drug exposure and potential efficacy or side effects for neuroactive compounds. Reported as % Unbound. Mouse Gastrocnemius Muscle Binding (MGMB):. This reflects the amount of drug free to act within skeletal muscle tissue, which is important for drugs targeting peripheral or muscular conditions (very important for the DM1 indication). Reported as % Unbound

While training the model, pay close attention to timing information. You are running in a virtual machine with ~32GB of RAM and ~8 CPUs and 1200 seconds per cell timeouts.

You must have predictions in a csv (same format as training) for the test compounds.

You should do a literature search (after dataset exploration, because chemistry tasks require some specific featurization, model architectures, data splitting, and training strategies). Although be careful about compute time!

You have been provided as attachments information about the ADMET properties and the context in which the data were collected. Use that to guide modeling decisions. If there are existing chemical informatics or machine learning models that are available to install as packages or download from accessible repositories, use them. However, be careful about your computational resource constraints, ability to access internet resources, and ability to install software in a sandboxed environment.

Dataset Description

The dataset contains the following files:
advancing-predictive-ADMET-modeling.pdf - a PDF with context about the dataset being modeled.
announcement-2-ExpansionRx-OpenADMET-blind-challenge.pdf - a PDF with context about the dataset being modeled.
expansion$_{\text{data}}$_test$_{\text{blinded}}$.csv - the compounds for which predictions must be made as SMILES strings.
expansion$_{\text{data}}$_train.csv - the training dataset with SMILES strings and the ADMET properties.
expansion$_{\text{data}}$_train$_{\text{raw}}$.csv - an alternative format of the training dataset in raw form.
Expansion-CDD-webinar.pdf - a PDF with context about the dataset being modeled.
OpenADMET-ExpansionRx-Blind-Challenge-about.pdf - a PDF with context about the dataset being modeled.
OpenADMET-ExpansionRx-Blind-Challenge-FAQ.pdf - a PDF with context about the dataset being modeled.

Summary of Discoveries

Discovery 1: Data sparsity, skewed distributions, and domain shift across ADMET endpoints

This work establishes a rigorous, end-to-end ADMET modeling pipeline on a sparse, heterogeneous, and distributionally skewed benchmark, showing that careful preprocessing and scaffold-aware validation yield strong baselines on five of nine endpoints. Although the held-out test set shows a complexity shift by SMILES length, targeted stress tests did not reveal statistically significant degradation in predictive accuracy for representative endpoints.

Discovery 2: Optimized single-task learning with combined fingerprints and physicochemical descriptors plus stacked ensembling

Single-task, tree-based models using 2048-bit, radius-2 Morgan fingerprints augmented with six simple 2D physicochemical descriptors provide the most accurate and reliable foundation for predicting nine diverse ADMET endpoints. A lightweight stacked ensemble that linearly reweights fingerprint- and descriptor-only base models with Ridge regression consistently improves performance, with the largest gains for Caco-2 Papp A>B, while multi-task neural baselines underperform on this sparse, heterogeneous dataset.

Discovery 3: Structural novelty and uncertainty quantification enable reliable deployment and targeted triage

This work shows that predictive error in ADMET models is systematically linked to structural novelty and endpoint-intrinsic variability, and that explicit, model-agnostic uncertainty estimation is required for reliable deployment under domain shift. Scalable novelty metrics and error meta-models produce well-calibrated, property-specific uncertainty signals that enable statistically validated accuracycoverage gains and targeted triage of risky predictions.

Discovery 4: Mechanistic validity and species-specific learning in microsomal clearance models

Microsomal clearance models trained on circular fingerprints learn chemically intuitive, nitrogen-centered metabolic liabilities and capture species differences, but human liver microsome (HLM) models are vulnerable to dataset-level confounding by bulk properties that can invert causal doseresponse tests. Adding explicit physicochemical descriptors (molecular weight, lipophilicity, and polarity) restores mechanistic behavior, eliminates a statistically significant reverse trend, and yields more plausible estimates.

Data sparsity, skewed distributions, and domain shift across ADMET endpoints

Summary

This work establishes a rigorous, end-to-end ADMET modeling pipeline on a sparse, heterogeneous, and distributionally skewed benchmark, showing that careful preprocessing and scaffold-aware validation yield strong baselines on five of nine endpoints. Although the held-out test set shows a complexity shift by SMILES length, targeted stress tests did not reveal statistically significant degradation in predictive accuracy for representative endpoints.

Background

Structureproperty modeling of ADMET endpoints is central to medicinal chemistry decision-making because lipophilicity, solubility, microsomal stability, permeability, and tissue-specific unbound fractions collectively determine exposure, efficacy, and safety risks. Industrial datasets typically exhibit high missingness, heterogeneous assay coverage, censoring at quantitation limits, and heavy-tailed measurement distributions. Robust cheminformatics pipelines therefore rely on appropriate transformations of skewed endpoints, chemically meaningful data splits, and model classes that can learn from limited, imbalanced labels while remaining resilient to domain shifts as chemistry evolves during lead optimization.

Results & Discussion

The benchmark comprises 5,326 training and 2,282 test molecules spanning nine ADMET endpoints, with 38.69% overall missingness and highly uneven coverage across assays (from 96.3% for kinetic solubility to 4.2% for mouse gastrocnemius muscle binding), alongside a small fraction of censored observations recorded with inequality modifiers in a raw file variant [r0]. Distributional analysis shows strong right-skew for HLM and MLM intrinsic clearance and for Caco-2 efflux ratio (e.g., HLM CLint skew=6.99, MLM CLint skew=4.19, efflux skew=5.90), and biologically coherent correlations including HLMMLM clearance (r=0.56), LogDMPPB (r=-0.69), and MBPBMGMB (r=0.90) that motivate multitask learning [r0]. The traintest partition exhibits a design-consistent domain shift: test SMILES strings are approximately 20% longer on average (57.8 vs 48.0), suggesting later-stage, structurally more elaborate compounds in the blind set [r0]. Censored values constitute 0.322.12% of available measurements, emphasizing the need for principled handling of limits of quantitation in model training [r0].

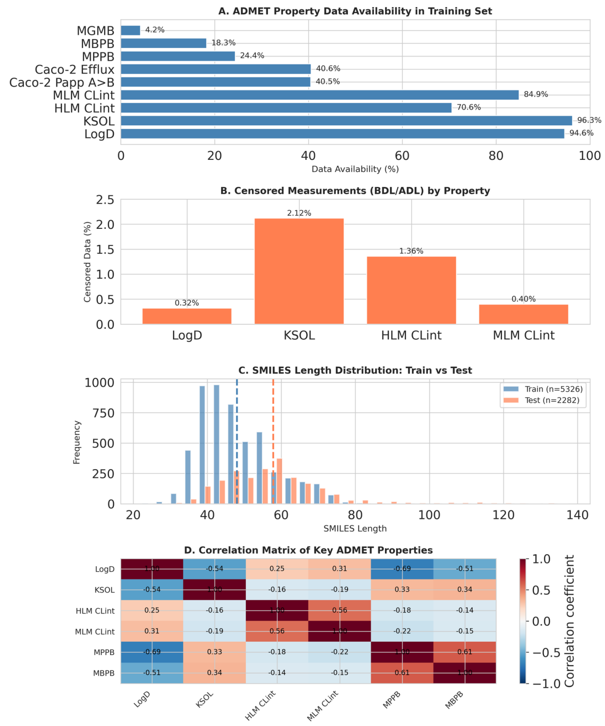

Figure 1: The ADMET benchmark dataset is characterized by data sparsity, a structural domain shift, and inter-property correlations. (A) Data availability in the training set varies widely across endpoints, from over 90% for LogD and KSOL to just 4.2% for MGMB. (B) A small fraction of measurements are censored, ranging from 0.32% to 2.12% for select properties. (C) A distributional shift is observed between the training and test sets, with test molecules having longer SMILES strings on average. (D) The correlation matrix reveals biologically coherent relationships, such as the positive correlation between HLM and MLM clearance. These dataset features highlight key challenges and opportunities for predictive modeling. (Source: [r0])

A single-task baseline trained independent LightGBM regressors on 2048-bit, radius-2 Morgan fingerprints, applying $\log_{10}$ transforms to the three most skewed endpoints (HLM_{CLint}, MLM_{CLint}, $Caco2_{Efflux}$) and using 5-fold BemisMurcko scaffold-based cross-validation to approximate generalization to novel chemotypes [r2]. This protocol yielded high performance (R^2 0.5) on five endpoints: LogD (R^2=0.804ś0.026), MLM_{CLint} ($\log_{10}$) (0.601ś0.011), KSOL (0.535ś0.023), HLM_{CLint} ($\log_{10}$) (0.511ś0.042), and $Caco2_{Efflux}$ ($\log_{10}$) (0.504ś0.094) [r2]. Three endpoints achieved moderate performance (0.3 $R^2 < 0.5$)MPPB (0.448ś0.031), MBPB (0.401ś0.075), and $Caco2_{Papp}$ (0.369ś0.079)while MGMB, with just 223 labels (4.2% coverage), was unstable and low-performing (R^2=0.282ś0.265) [r2]. Fold sizes were balanced (~20% per fold), with 2,351 unique scaffolds identified across the dataset, and early stopping was used to regularize training on each property-specific labeled subset [r2].

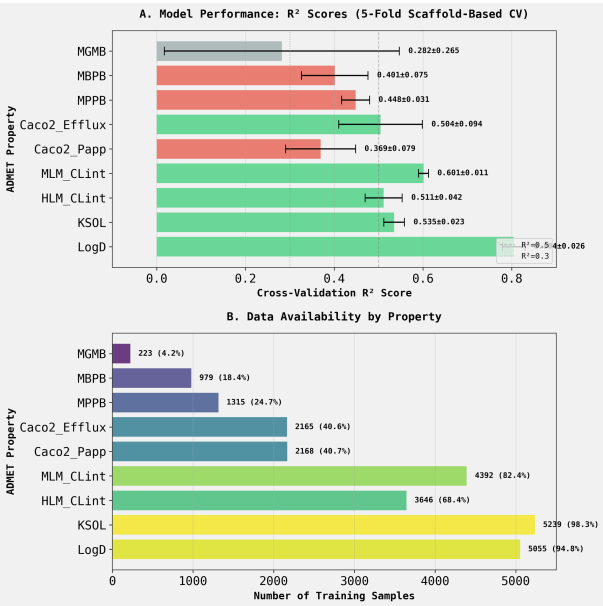

Figure 2: Single-task model performance is strongly associated with data availability across ADMET endpoints. (A) Mean coefficient of determination (R^2) from 5-fold scaffold-based cross-validation for individual LightGBM models, with error bars denoting standard deviation. (B) The number and percentage of training samples available for each of the nine ADMET properties. Models achieve strong predictive performance (R^2 0.5) on five endpoints with high data coverage, while performance is substantially lower for sparse endpoints such as MGMB, MBPB, and MPPB. (Source: [r2])

Two design choices were especially consequential. First, $\log_{10}$-transforming HLM/MLM clearance and efflux ratio normalized the heavy right tails and was associated with all three transformed endpoints exceeding the R^2 0.5 threshold, in line with the distributional diagnostics from the exploratory analysis [r0, r2]. Second, data availability dominated performance: properties with >2,000 labels averaged $R^2 \approx 0.554$, whereas those with <1,000 labels averaged $R^2 \approx 0.342$, rationalizing the difficulty of MGMB and suggesting that correlated, better-covered endpoints could serve as auxiliary tasks in a multitask setting [r2]. Notably, $Caco2_{Papp}$ remained comparatively challenging (R^2=0.369) despite ~2,168 measurements, implying either higher intrinsic assay variance or the need for features beyond 2D circular fingerprints for permeability (e.g., physicochemical, 3D, or conformational descriptors) [r2].

Because the blind test set contains longer SMILES, a targeted robustness experiment assessed whether prediction error increases disproportionately on structurally complex molecules. Using 5-fold cross-validation with KFold shuffling, a 95th-percentile SMILES-length threshold (computed within each training fold) defined an out-of-distribution group comprising only 36% of validation molecules per fold; models were LightGBM ensembles over fingerprints-only and fingerprints+descriptors and evaluated by MAE with paired t-tests at α=0.05 [r22]. For LogD, MAE increased by 14.5% on complex molecules (p=0.059), while for MPPB, MAE decreased by 14.2% (p=0.416), with both effects not statistically significant and characterized by high variance due to small out-of-distribution sample sizes [r22]. Together with the observed 20% SMILES-length increase from train to test, these results indicate a measurable but modest complexity shift that the current featuremodel stack largely tolerates, while also highlighting that SMILES length is a crude proxy and that richer complexity metrics (e.g., stereocenter count, ring systems, or rarity of functional groups) are likely needed to more sensitively capture domain shift risk [r0, r22]. Strategically, the biological correlations and data sparsity patterns motivate multitask learning to lift low-label endpoints (e.g., MBPB, MGMB), incorporation of physicochemical and 3D features to improve permeability prediction, and exploration of censored-regression variants and distribution-shift mitigation to better align models with assay realities and temporal evolution of chemistry [r0, r2].

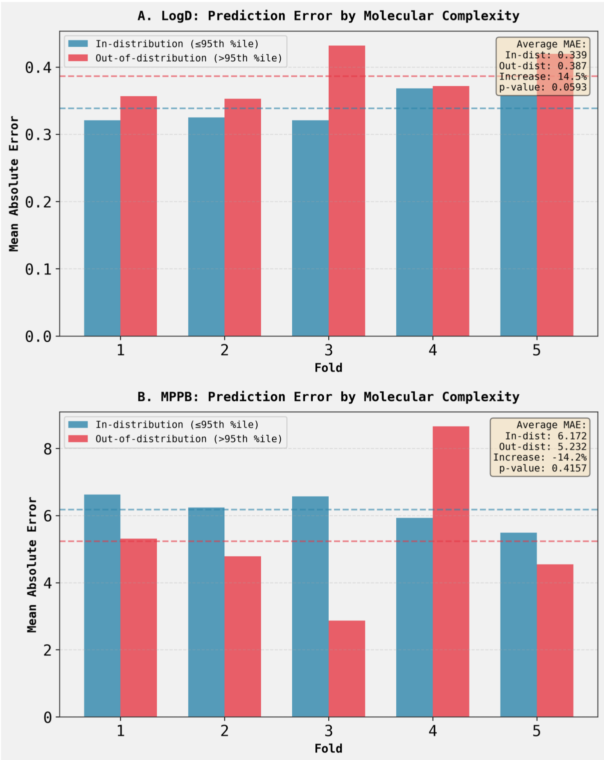

Figure 3: Model performance does not significantly degrade for molecules of high structural complexity. Mean Absolute Error (MAE) across five cross-validation folds for (A) LogD and (B) MPPB models, stratified by whether a molecule's complexity is within the 95th percentile (in-distribution) or above it (out-of-distribution). The average MAE difference between the two complexity strata is not statistically significant for either endpoint ($p > 0.05$), suggesting the models are robust to this type of domain shift. (Source: [r22])

Trajectory Sources

Trajectory r0: This dataset represents the ExpansionRx-OpenADMET Blind Challenge with 5,326 training molecules and 2,282 test molecules containing 9 ADMET properties with variable data availability (4.2%-96.3%) and significant distributional challenges including high missingness, right skewness in clearance measur...

Trajectory r2: Training independent LightGBM models with Morgan fingerprints on log-transformed skewed properties and scaffold-based cross-validation yielded a strong baseline with 5 of 9 ADMET properties achieving R^2 0.5, with LogD performing best at $R^2 = 0.804$ ś 0.026.

Trajectory r22: The hypothesis that structurally complex molecules (SMILES length >95th percentile) have at least 25% higher prediction error than less-complex counterparts is not supported for either LogD or MPPB properties, with LogD showing a 14.5% increase (p=0.059, not significant) and MPPB showing a -14.2% de...

Optimized single-task learning with combined fingerprints and physicochemical descriptors plus stacked ensembling

Summary

Single-task, tree-based models using 2048-bit, radius-2 Morgan fingerprints augmented with six simple 2D physicochemical descriptors provide the most accurate and reliable foundation for predicting nine diverse ADMET endpoints. A lightweight stacked ensemble that linearly reweights fingerprint- and descriptor-only base models with Ridge regression consistently improves performance, with the largest gains for Caco-2 Papp A>B, while multi-task neural baselines underperform on this sparse, heterogeneous dataset.

Background

Predicting absorption, distribution, metabolism, excretion, and toxicity (ADMET) from structure remains a central challenge in drug discovery because assays probe distinct biophysical and biological mechanisms, the data are sparse and noisy, and generalization must hold across novel chemical scaffolds. Lipophilicity (LogD), solubility (kinetic solubility), microsomal clearance (HLM and MLM), intestinal permeability (Caco-2 Papp and efflux), and protein/tissue binding (MPPB, MBPB, MGMB) collectively determine exposure and efficacy, yet they differ in data availability, scale, and statistical properties. Robust modeling requires feature representations that capture local structural context and global physicochemical constraints, validation schemes that reflect scaffold novelty, and post hoc constraints that enforce biological plausibility.

Results & Discussion

The dataset comprises 5,326 training molecules and 2,282 test molecules spanning nine ADMET properties with highly uneven availability (4.296.3% measured), extensive missingness (38.69% of all values), right-skewed distributions for clearance and efflux, and a realistic temporal/domain shift (test SMILES are ~20% longer; mean 57.8 vs 48.0), alongside a small fraction of censored measurements in the raw files, all of which complicate model training and evaluation [r0]. A rigorous single-task baseline trained LightGBM regressors on 2048-bit, radius-2 Morgan fingerprints with log10 transformations for HLM CLint, MLM CLint, and efflux to mitigate skewness, using five-fold scaffold-based cross-validation; this achieved R^2 0.5 for five endpoints, led by LogD (R^2 = 0.804 ś 0.026), with strong results for MLM CLint (0.601 ś 0.011), HLM CLint (0.511 ś 0.042), KSOL (0.535 ś 0.023), and Caco-2 efflux (0.504 ś 0.094), while permeability (Caco-2 Papp) and binding properties were moderate and MGMB exhibited high variance due to limited samples (N = 223) [r2].

Augmenting fingerprints with six 2D physicochemical descriptors (molecular weight, TPSA, hydrogen-bond donors and acceptors, MolLogP, and rotatable bonds) improved accuracy across all endpoints, confirming that global physicochemical context adds non-redundant signal to local structural features. Systematic comparison across the nine properties showed that the combined feature set was numerically best for all endpoints and significantly better than both fingerprints-only and descriptors-only for six (and significantly better than one alternative for two), with the largest relative gains for Caco-2 Papp A>B (+0.1211 R^2; +27.2%), MPPB (+0.1110), and MBPB (+0.1030) [r62]. A focused ablation on HLM clearance reinforced this synergy: descriptors-only (R^2 = 0.295 ś 0.026) underperformed fingerprints-only (0.505 ś 0.017), and the combined model achieved the highest performance (0.531 ś 0.024), with improvements over both feature baselines statistically significant (combined vs fingerprints p = 0.022; combined vs descriptors p < 1 Œ 10^{-4}) [r42].

Building on this complementarity, a stacked ensemble that linearly reweights the predictions of fingerprint-only and descriptor-only LightGBM models with a Ridge meta-learner in a fully nested cross-validation setup improved per-

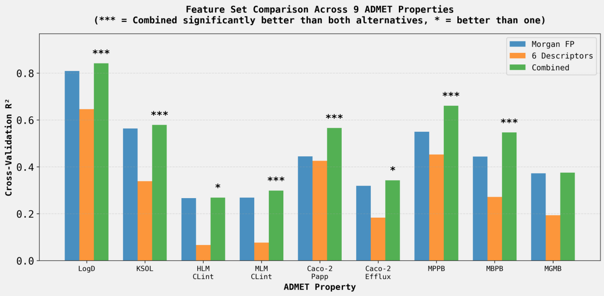

Figure 4: Combining Morgan fingerprints with six physicochemical descriptors consistently improves model performance across nine ADMET properties. The bar chart compares the cross-validation R^2 for models trained using Morgan fingerprints only, descriptors only, or a combined feature set. The results demonstrate that global physicochemical properties provide predictive information that is non-redundant with the local structural features captured by fingerprints. (Source: [r62])

formance for all nine endpoints and achieved statistical significance for five. The strongest gain was for Caco-2 Papp A>B (R^2 0.466 0.534; +14.5%; p = 0.0004), with significant but smaller increments for HLM CLint (0.506 0.519; p = 0.039), MLM CLint (0.634 0.643; p = 0.018), LogD (0.842 0.854; p = 0.0002), and KSOL (0.578 0.586; p = 0.034), and consistent, albeit non-significant, improvements for the remaining properties [r63]. Analysis of the meta-learner coefficients showed that fingerprints generally dominate the linear combination (5283% weight), but permeability and binding receive more balanced contributions (e.g., Caco-2 Papp A>B 52.4% fingerprint/47.6% descriptor; MPPB 61.9%/38.1%), aligning with the mechanistic expectation that both local structural motifs and bulk properties govern passive diffusion and protein/tissue partitioning [r84].

In contrast, multi-task neural baselines underperformed. Independent Random Forests surpassed a multi-task feedforward network by a wide margin (mean R^2 0.508 vs 0.386), with large advantages for lipophilicity, clearance, and efflux, and only slight, high-variance benefits for the most data-sparse endpoints; these single-task models were consequently used to generate test-set predictions in the required format [r4]. A multi-task Chemprop D-MPNN further deteriorated performance across all nine properties, including drastic deficits versus the LightGBM baseline for LogD (R^2 0.213 vs 0.804), MLM

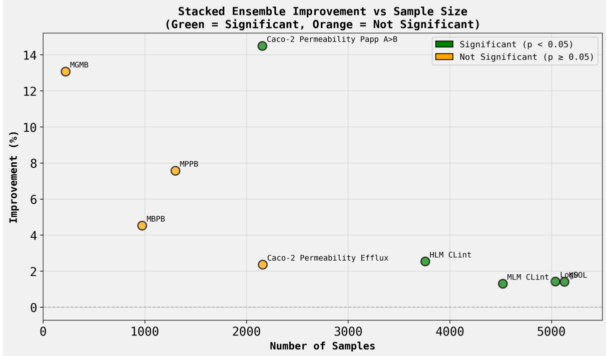

Figure 5: Stacked ensembling consistently improves predictive performance over the single best model across all ADMET endpoints. The plot shows the percentage improvement from the stacked ensemble as a function of the number of samples for each endpoint, colored by statistical significance (green, $p < 0.05$; orange, p 0.05). While the largest percentage gains occur for endpoints with smaller sample sizes, the stacked ensemble provides the most impactful and statistically significant improvement for Caco-2 permeability (Papp A>B). (Source: [r63])

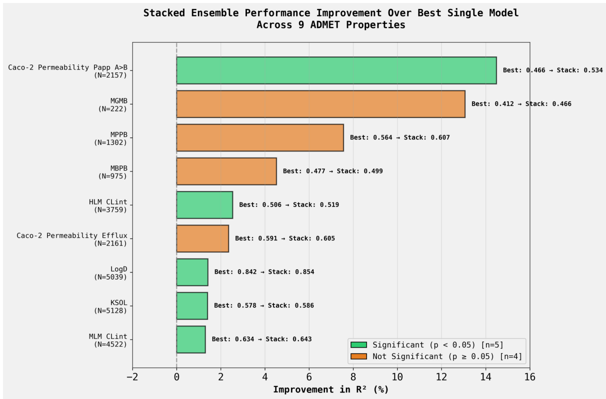

Figure 6: A stacked ensemble model consistently improves predictive performance over the best single-task model across nine ADMET properties. Bars show the percentage improvement in the test set coefficient of determination (R^2) for the stacked ensemble relative to the best single model, with statistically significant improvements ($p < 0.05$) highlighted in green. The ensemble provides the largest benefit for Caco-2 permeability (Papp A>B), with a statistically significant increase in R^2 of nearly 15%. (Source: [r63])

CLint (0.074 vs 0.601), and Caco-2 Papp (0.063 vs 0.369), consistent with underfitting/negative transfer in the face of heterogeneous tasks, high missingness, and limited compute for tuning and early stopping [r0, r2, r5].

Feature-level analyses confirm that these models learn chemically meaningful rules and illuminate why fusing global descriptors with local fingerprints is beneficial. For KSOL, amide/urea and

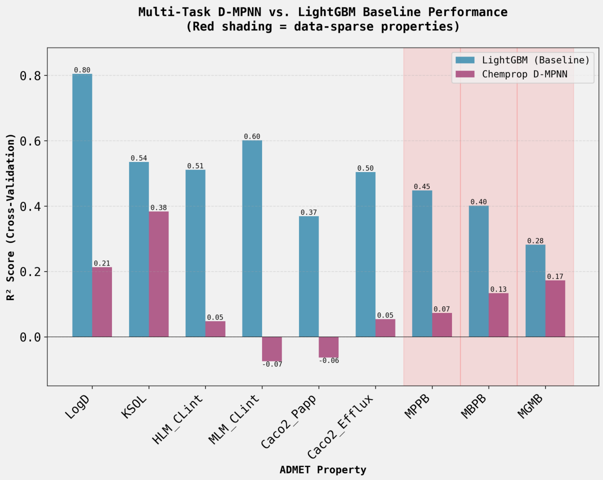

Figure 7: The single-task LightGBM baseline consistently outperforms a multi-task D-MPNN model across nine ADMET endpoints. The bar plot compares the cross-validation R^2 score for the LightGBM model (blue) against the D-MPNN (magenta) for each prediction task, with data-sparse properties highlighted in red. These results establish that a single-task learning framework is more effective than a multi-task approach for this sparse and heterogeneous dataset. (Source: [r5])

carbonyl-containing contexts increased solubility substantially (e.g., +44 to +103 μg/mL), while certain nitrogen heterocycles decreased it; for LogD, basic amines reduced lipophilicity (0.39), whereas aromatic nitriles and fluorinated aromatics increased it (+0.55 to +1.24), capturing intuitive structureproperty relationships and the classic lipophilicitysolubility trade-off [r16]. Clearance models most influential bits map to well-known Phase I liabilities (N-dealkylation, aromatic/heterocycle oxidation), with a notable species difference for a pyrazole-associated bit that was 23Œ more important in MLM than HLM, highlighting biologically plausible interspecies metabolism differences [r24]. Finally, enforcing biological plausibility post hoc by clamping binding predictions at zero removed 158 negative values and modestly improved performance ($\Delta R^2 = +0.0011$ to $+0.0083$; ΔRMSE uniformly favorable), supporting the routine use of this simple constraint for protein/tissue binding endpoints [r9]. Together, these results establish single-task tree models with combined fingerprints and physicochemical descriptors, optionally stacked via Ridge reweighting, as a robust, interpretable, and computationally efficient strategy for this open ADMET challenge [r2, r62, r63].

Trajectory Sources

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Trajectory r4: Single-task Random Forest models outperformed multi-task neural networks in cross-validation (average $R^2 = 0.508$ vs 0.386), and were retrained on the entire training dataset to generate predictions for all 9 ADMET properties across 2,282 test molecules, saved to predictions.csv.

Trajectory r5: The multi-task Directed Message Passing Neural Network (D-MPNN) implemented via Chemprop significantly underperformed compared to the single-task LightGBM baseline, with all 9 ADMET properties showing worse R^2 scores and the data-sparse properties (MGMB, MBPB, MPPB) experiencing severe performance d...

Trajectory r9: Post-prediction clamping (max(0, prediction)) successfully eliminates all biologically implausible negative predictions for binding properties (MPPB, MBPB, MGMB) while actually improving performance ($\Delta R^2 =$ +0.0011 to +0.0083), confirming the hypothesis that clamping does not degrade performance by m...

Trajectory r16:
Analysis Complete: Chemical Substructures Driving KSOL and LogD Predictions
Summary
The analysis successfully identified and visualized the top 5 most important Morgan fingerprint bits for predicting both Kinetic Solubility (KSOL) and Lipophilicity (LogD) using LightGBM models trained on 2...

Trajectory r24: All 10 of the most important Morgan fingerprint bits for HLM CLint and MLM CLint clearance prediction correspond to known Phase I metabolic liabilities, predominantly nitrogen-containing substructures susceptible to N-dealkylation, aromatic oxidation, and N-oxidation.

Trajectory r42: A model trained on six physicochemical descriptors alone achieved $R^2 = 0.295$ ś 0.026 for HLM clearance prediction, which is significantly lower than Morgan fingerprints alone ($R^2 = 0.505$ ś 0.017, $p < 0.0001$), and the combined model achieved $R^2 = 0.531$ ś 0.024, significantly outperforming both finger...

Trajectory r62:
SYSTEMATIC FEATURE SET COMPARISON: OPTIMAL CHOICE FOR 9 ADMET PROPERTIES
FINAL CONSOLIDATED SUMMARY
The systematic comparison of three feature sets (Morgan fingerprints-only, 6 physicochemical descriptors-only, and combined features) across all 9 ADMET properties using 5-fold cross-validat...

Trajectory r63: A stacked ensemble using Ridge regression to combine fingerprint-only and descriptor-only LightGBM base models improved performance over the best single model for all 9 ADMET properties, with statistically significant improvements ($p < 0.05$) for 5 properties including Caco-2 Papp A>B (+14.5% improve...

Trajectory r84: The Ridge meta-learner coefficients reveal that fingerprint-based models consistently dominate across all 9 ADMET properties (52-83% weight), with the most balanced weighting for Caco-2 permeability (52% fingerprint / 48% descriptor) and the strongest fingerprint dominance for metabolic clearance an...

Structural novelty and uncertainty quantification enable reliable deployment and targeted triage

Summary

This work shows that predictive error in ADMET models is systematically linked to structural novelty and endpoint-intrinsic variability, and that explicit, model-agnostic uncertainty estimation is required for reliable deployment under domain shift. Scalable novelty metrics and error meta-models produce well-calibrated, property-specific uncertainty signals that enable statistically validated accuracycoverage gains and targeted triage of risky predictions.

Background

Accurate machine learning prediction of ADMET endpoints underpins modern medicinal chemistry, yet deployment often fails when test compounds diverge from training chemical space or when assays exhibit high intrinsic variability. Structural novelty inflates extrapolation error, while aleatoric noise limits achievable accuracy even with abundant data. Consequently, uncertainty quantificationideally grounded in both chemical similarity and property-specific mechanismsis essential to prioritize high-confidence decisions, flag risky compounds for confirmatory assays, and systematically improve accuracy by trading coverage for confidence.

Results & Discussion

The test set exhibits pronounced domain shift that can be prospectively identified with a continuous novelty score computed from Morgan fingerprints as the minimum frequency of active bits in the training set. Test molecules have significantly lower novelty than training (KolmogorovSmirnov statistic=0.3277, $p < 1.90$ Œ 10^{-152}, Cohens d=0.578), with 35.7% falling below a low-confidence threshold (10th percentile of training) and 4.6% containing never-seen substructures (novelty=0), directly signaling elevated risk at deployment time [r32]. This novelty signal is consequential: across two representative endpoints, prediction error is negatively correlated with novelty (Spearman ρ = -0.1597 for LogD, p = 3.72 Œ 10^{-30}; ρ = -0.1017 for MPPB, p = 2.36 Œ 10^{-4}), and compounds in the lowest-novelty quartile have 47% and 43% higher mean absolute error than the highest-novelty quartile for LogD and MPPB, respectively, demonstrating a monotonic link between substructural rarity and error [r29]. Critically, the largest errors are not confined to a few underrepresented scaffolds: in HLM CLint, the top 5% error compounds are more novel (t = -5.279, p = 1.37 Œ 10^{-7}) yet span 158 unique scaffolds (84.0% unique), with the most common scaffold accounting for only 3.7% of failures, refuting a few bad scaffolds hypothesis and underscoring broad novelty-driven risk [r43].

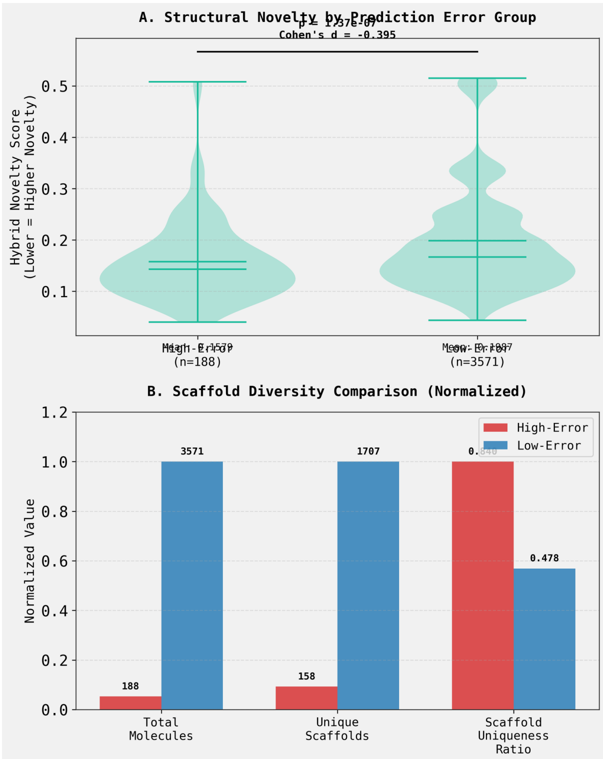

Figure 8: High-error predictions are associated with increased structural novelty and high scaffold diversity. (A) Violin plots show that molecules in the high-error prediction group (n=188) have significantly lower novelty scores, indicating greater novelty, compared to the low-error group (n=3571). (B) The high-error group has a much higher scaffold uniqueness ratio (0.840 vs 0.478), showing greater structural diversity relative to its size. This high diversity indicates that prediction risk is broadly distributed across many structures rather than being confined to a few problematic scaffolds. (Source: [r43])

A property-agnostic error meta-model that integrates fingerprints, the primary prediction, and novelty produces strong uncertainty estimates and remains effective even in extreme data scarcity. For MGMB (N=222), predicted and true absolute errors correlate at r = 0.5132 (p = 1.36 Œ 10^{-15}), exceeding analogous performance on data-rich properties and demonstrating that learnable error structure can dominate sample-size limitations [r57]. More broadly, the full meta-model approximately doubles performance over a novelty-only baseline across both data-rich LogD and data-sparse MPPB (Δr $\approx$ +0.42; LogD r: 0.42470.8502, MPPB r: 0.41560.8222), confirming that structural features and prediction magnitude add complementary uncertainty signal beyond novelty alone [r68]. To scale novelty features, approximate nearest neighbor search with a Jaccard metric is provably equivalent to 1 Tanimoto for binary fingerprints and is 5.9Œ faster than brute-force similarity, completing novelty computation for N=5,039 in 1.83 s and yielding perfect agreement with the Tanimoto method when both use the nearest neighbor (Spearman $\rho = 1.0000$), enabling practical deployment at scale [r59].

Conformal prediction further clarifies that uncertainty reflects both epistemic limits and endpoint-intrinsic (aleatoric) variability. Across nine ADMET endpoints, the hypothesis that 95% interval width inversely tracks training size is not supported: MGMB (N=222) has much narrower average intervals than KSOL (32.4 vs 358.7 units), and interval width shows no significant inverse correlation with sample size (Spearman ρ = 0.267, p = 0.488), with relative widths showing only a weak, non-significant inverse trend (ρ = -0.350, p = 0.356) [r10]. These findings indicate that assay variability and modeling difficulty can outweigh sample size, mandating property-aware interpretation of absolute interval widths. The analysis delivers point predictions and 95% conformal intervals for 2,282 test molecules in test$_{\text{predictions__}}$with$_{\text{uncertainty}}$.csv, with coverage-valid, scale-aware uncertainty that complements novelty and meta-model estimates [r10].

Uncertainty also arises from property-specific mechanisms beyond novelty, creating combination-driven blind spots that standard multivariate outlier metrics miss. For KSOL,

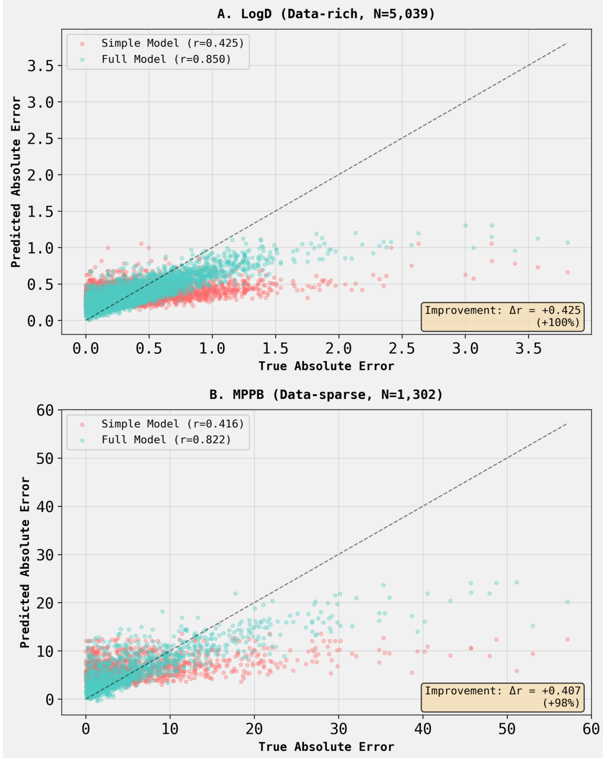

Figure 9: A full error meta-model substantially improves the prediction of true absolute error compared to a simple baseline. Predicted absolute error is plotted against true absolute error for a simple model (red) and a full model (cyan) on (A) the data-rich LogD endpoint and (B) the data-sparse MPPB endpoint. The full model's predictions cluster tightly around the identity line (y=x) and achieve a Pearson correlation (r) nearly double that of the simple model, indicating a significantly more accurate and well-calibrated error signal. (Source: [r68])

high-uncertainty predictions among structurally typical molecules are enriched 4.04Œ for a polar + heavy patternhigh TPSA or many H-bond acceptors together with high molecular weight and relatively low lipophilicityand a targeted conflict score combining descriptor interactions strongly separates these cases (t = 14.972, p < 0.001), whereas Mahalanobis distance does not (r $\approx$ 0.04), revealing directional physicochemical conflicts as the driver [r76]. In contrast, MGMB exhibits the opposite pattern: high uncertainty concentrates in lighter, low-lipophilicity molecules (odds ratio=5.32 for MolWt < 400, p = 0.005; odds ratio=3.41 for MolLogP < 2.5, p = 0.038), with elevated conflict scores driven primarily by the lipophilicity term, demonstrating that uncertainty mechanisms are endpoint-specific [r80]. Practically, these calibrated uncertainty estimates enable statistically significant accuracycoverage gains across diverse properties: at 90% coverage, RMSE drops by 7.6% (LogD), 38.7% (HLM CLint), and 17.5% (MPPB), and at 80% coverage by 10.8%, 50.7%, and 26.1%, respectively (all p $<$ 0.001 by permutation), validating targeted triage as an operational strategy for reliable deployment [r83]. Final deliverables include predictions with confidence for 2,282 test molecules in final_predictions_with_confidence.csv, supporting risk-aware decision-making under pronounced test-set shift [r32, r83].

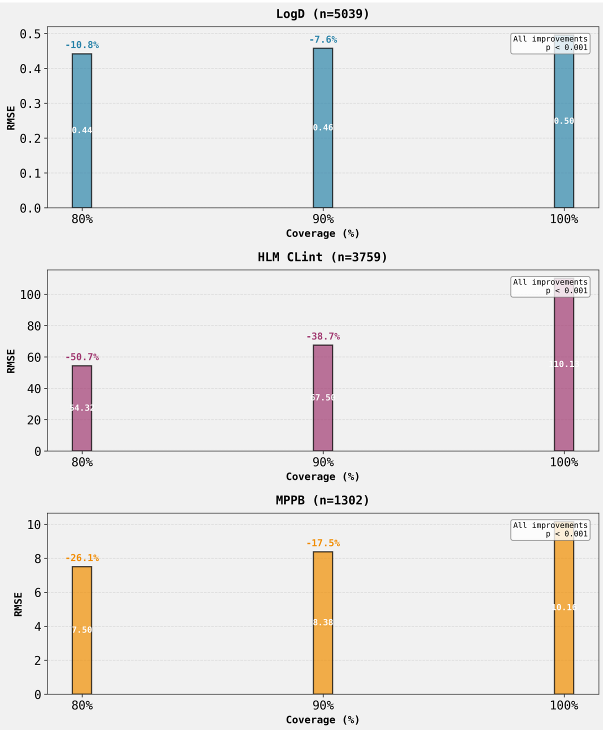

Figure 10: Targeted triage of uncertain predictions enables statistically significant accuracy-coverage gains. The bar plots show the Root Mean Square Error (RMSE) for three ADMET endpoints (LogD, HLM CLint, and MPPB) at decreasing levels of test set coverage. By deferring the 10% and 20% most uncertain predictions to achieve 90% and 80% coverage, respectively, model error is substantially reduced across all endpoints (all improvements p $<$ 0.001), demonstrating a practical trade-off for reliable model deployment. (Source: [r83])

Trajectory Sources

Trajectory r10: The hypothesis that conformal prediction interval width is inversely correlated with training data amount is NOT CONFIRMED: MGMB (N=222) has an average 95% prediction interval width of 32.4 units, which is 11 times SMALLER than KSOL (N=5,128) with 358.7 units, opposite to the predicted 5-fold larger...

Trajectory r29:
Answer
The continuous Novelty Score, defined as the minimum frequency of constituent Morgan fingerprint bits in the training set, shows **statistically significant negative correlations** with prediction error for both LogD and MPPB properties, confirming the research hypothesis.
Key Quant...

Trajectory r32: The blind test set contains 815 molecules (35.7%) that fall below the low-confidence threshold, indicating substantial structural novelty compared to the training set, with 105 molecules (4.6%) containing substructures never observed in training (Novelty Score = 0).

Trajectory r43:
Analysis Complete: HLM Clearance Prediction Error Characterization
PRIMARY FINDINGS
Hypothesis Testing Results: 1. **SUPPORTED**: High-error molecules (top 5% by prediction error) have significantly higher structural novelty than low-error molecules 2. **REFUTED**: High-error molec...

Trajectory r57: The error-prediction meta-model framework successfully achieved a statistically significant correlation (r = 0.5132, p $<$ 0.0001) between predicted and true absolute errors for the extremely data-sparse MGMB property (N=222), rejecting the hypothesis that the framework would fail under severe data sc...

Trajectory r59: The ANN-based novelty score using k=1 (nearest neighbor) is computationally feasible for large datasets (1.83 seconds for N=5,039, well below 1200s limit) and achieves perfect correlation (Spearman ρ = 1.0000, p $<$ 0.001) with the Tanimoto-based method on the MPPB dataset, demonstrating mathematical ...

Trajectory r68: The full error-prediction meta-model (using fingerprints, novelty score, and primary prediction) significantly outperforms a simple novelty-only model, achieving approximately 100% improvement in performance (Δr $\approx$ +0.42) across both data-rich (LogD, N=5,039) and data-sparse (MPPB, N=1,302) propertie...

Trajectory r76: High uncertainty predictions for KSOL in structurally typical molecules are driven by specific conflicting physicochemical property combinationshigh molecular weight combined with high polarity (TPSA) and low lipophilicity (LogP)rather than general descriptor space sparsity as measured by Mahalano...

Trajectory r80: The "Polar + Heavy" physicochemical conflict pattern identified in KSOL does not generalize to MGMB; instead, MGMB exhibits a reverse pattern where high uncertainty in structurally typical molecules is driven by low lipophilicity (low MolLogP) combined with lighter molecular weight.

Trajectory r83: The uncertainty estimates demonstrate exceptional practical utility, with statistically significant RMSE reductions of 7.6-50.7% at 90% coverage and 10.8-50.7% at 80% coverage across three representative ADMET properties (LogD, HLM CLint, and MPPB), all with $p < 0.001$.

Mechanistic validity and species-specific learning in microsomal clearance models

Summary

Microsomal clearance models trained on circular fingerprints learn chemically intuitive, nitrogen-centered metabolic liabilities and capture species differences, but human liver microsome (HLM) models are vulnerable to dataset-level confounding by bulk properties that can invert causal doseresponse tests. Adding explicit physicochemical descriptors (molecular weight, lipophilicity, and polarity) restores mechanistic behavior, eliminates a statistically significant reverse trend, and yields more plausible estimates.

Background

High-throughput microsomal clearance assays are a cornerstone of medicinal chemistry optimization, linking molecular structure to in vitro hepatic intrinsic clearance and enabling early estimation of in vivo clearance. Modern machine learning models, particularly gradient-boosting trees trained on circular fingerprints, promise accurate and interpretable predictions by associating substructures with known metabolic pathways. However, ADMET endpoints are also governed by bulk physicochemical properties such as lipophilicity, molecular size, and polar surface area, which co-vary with substructural motifs and can confound structureactivity relationships if not explicitly controlled. Disentangling specific metabolic liabilities from these background trends is essential for mechanistic validity, cross-species translation, and trustworthy guidance for medicinal chemistry.

Results & Discussion

Fingerprint models learned biochemically plausible clearance rules and species differences. In single-task LightGBM regressions on log-transformed clearance, HLM and mouse liver microsome (MLM) models reached R^2 of 0.729 and 0.785, respectively, using 2048-bit, radius-2 Morgan fingerprints [r24]. Strikingly, 10/10 of the most important fingerprint bits for both HLM and MLM corresponded to nitrogen-containing Phase I liabilities (N-dealkylation, aromatic oxidation, N-oxidation), with clear medicinal chemistry interpretability [r24]. A pronounced species signal emerged for pyrazole: a Morgan bit mapping to a pyrazole N-heterocycle exhibited 23Œ higher gain importance in MLM than HLM (gain 5271 vs 232), consistent with species-specific differences in CYP-mediated heterocycle metabolism [r24]. These results indicate that fingerprint-only models can capture meaningful structuremetabolism patterns and interspecies differences, at least at the level of local substructures [r24].

Despite high apparent accuracy, causal tests revealed a mechanistic failure in the HLM fingerprint-only model. A controlled in silico perturbation study replaced tertiary amine nitrogens with carbon to block known liability sites in molecules containing at least two tertiary amines. Contrary to biochemical expectation, back-transformed HLM predictions increased from Original (two amines) to Mod1 (one amine) to Mod2 (zero amines): 12.29 17.59 23.57 ţL/min/mg (n=15), with a significant Mod1Mod2 increase (t=-3.6451, one-tailed p=0.0013), indicating a reverse doseresponse where removing liabilities increased predicted clearance [r33]. The LightGBM models training R^2 (0.7287) underlines that standard fit metrics can mask mechanistic invalidity when models exploit confounded associations rather than causal structure [r33].

The reverse trend was traced to systematic physicochemical shifts induced by amine blocking and was eliminated by adding explicit descriptors. Each blocked tertiary amine increased lipophilicity and reduced polarity (ΔMolLogP $= +1.23$ per amine, $r_s = 0.715$, $p = 3.5Œ10^{-8}$; ΔTPSA $= 3.24$ Å^2 per amine, $r_s = 0.193$, $p = 0.20$), with a small mass decrease (ΔMolWt $=$ 0.99 Da, $r_s = 0.103$, $p = 0.50$) [r36]. In the fingerprints-only model, these property changes produced a significant reverse Mod1Mod2 increase (+6.53 ţL/min/mg; $t = 2.312$, $p = 0.0183$). Augmenting the model with MolWt, MolLogP, and TPSA reversed the trend (OriginalMod2: 18.01 ţL/min/mg, one-tailed p =

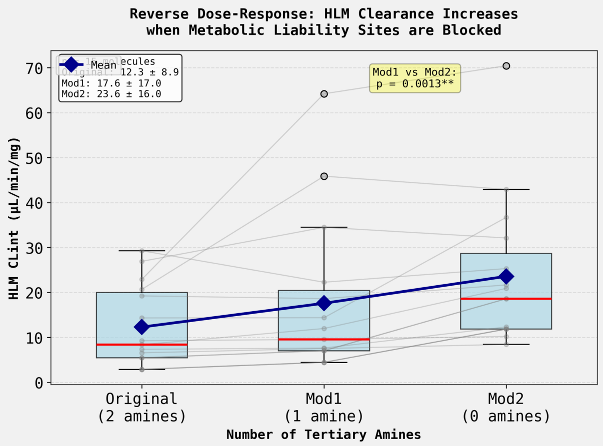

Figure 11: Blocking known metabolic liability sites paradoxically increases predicted HLM clearance in the fingerprint-only model. Predicted HLM intrinsic clearance (CLint) is shown for 15 original molecules containing two tertiary amines and their modified analogs where one (Mod1) or both (Mod2) liability sites are blocked. The mean predicted clearance (blue line) increases as liability sites are removed, with the increase from Mod1 to Mod2 being statistically significant (p=0.0013). This counterintuitive reverse dose-response demonstrates a mechanistic failure of the model when tested on a controlled chemical series. (Source: [r33])

0.077) and abolished the reverse effect (reverse test p = 0.92), constituting a 329.4% net correction (25.87 ţL/min/mg) toward the biochemically correct direction [r36]. Thus, explicit descriptors decoupled substructural liability signals from background property shifts, restoring mechanistic plausibility without materially changing the modeling framework [r36].

Orthogonal correlation analyses reinforced that HLM and MLM encode polarityclearance relationships differently when trained on fingerprints alone. In HLM, the top 20 fingerprint features had low mean absolute correlation with TPSA (0.1500) and showed no systematic relation between their TPSA correlation and clearance correlation (r = 0.122, p = 0.608), matching the very weak direct TPSAHLM clearance correlation (r = 0.046, p = 0.005) [r55]. In contrast, MLMs top features exhibited higher TPSA association (mean |r| = 0.2082) and a strong, mechanistically correct inverse relationship between TPSA correlation and clearance correlation (r = 0.771, p = 0.0001), consistent with a robust negative TPSAMLM clearance correlation (r = 0.352, p = 6.06Œ10^{-132}) [r55]. When TPSA and other descriptors were provided explicitly, TPSA-correlated fingerprints lost far more importance in HLM (76.4%) than MLM (15.5%), indicating that HLMs fingerprint bits were weak, confounded proxies for polarity that the explicit descriptor supplanted [r55]. Together, these results explain why HLM fingerprint-only models were prone to reverse causal responses while MLM models already captured the correct polarity trend [r55].

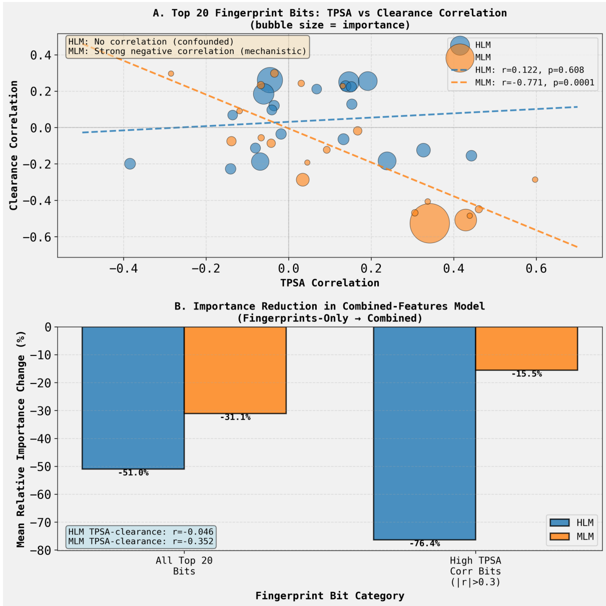

Figure 12: The human liver microsome (HLM) fingerprint model learns a confounding correlation with polarity that is not present in the mouse (MLM) model. (A) For the top 20 fingerprint bits from fingerprint-only models, clearance correlation is plotted against topological polar surface area (TPSA) correlation. The MLM model (orange) shows a significant negative trend (r=-0.771), while the HLM model (blue) shows no correlation; bubble size represents feature importance. (B) Mean relative reduction in the importance of these fingerprint bits after adding explicit physicochemical features. The importance of HLM bits highly correlated with TPSA (|r|>0.3) drops by 76.4%, much more than for MLM bits. These results suggest the HLM model uses fingerprint bits as confounded proxies for bulk polarity, an artifact corrected by including explicit descriptors. (Source: [r55])

Finally, matched-pair analyses clarified which properties must be controlled to remove systematic bias linked to tertiary amines and cautioned against single-variable matching. Using the Morgan bit corresponding to a tertiary amine as the case label, TPSA-only nearest-neighbor matching paradoxically increased bias in the combined-features model, while multivariate matching across TPSA, MolWt, and

MolLogP reduced the casecontrol error gap to the boundary of non-significance (fingerprints-only: mean difference 0.0688 log units, $t = 2.06$, $p = 0.040$; combined: 0.0654, $t = 1.95$, $p = 0.052$) [r56]. In the full dataset, tertiary aminebearing molecules had lower TPSA (2.50 $Å^2$), higher MolWt (+28.98 Da), slightly higher MolLogP (+0.08), and markedly higher true clearance (+0.595 log units) than controls, underscoring why TPSA-only control is insufficient [r56]. These results show that model attributions from single instances and naive matching can mislead; systematic matched-pair designs that control key covariates are required to validate learned rules and to ensure that descriptor augmentation corrects, rather than amplifies, confounding [r56].

Trajectory Sources

Trajectory r24: All 10 of the most important Morgan fingerprint bits for HLM CLint and MLM CLint clearance prediction correspond to known Phase I metabolic liabilities, predominantly nitrogen-containing substructures susceptible to N-dealkylation, aromatic oxidation, and N-oxidation.

Trajectory r33: The HLM clearance model does not exhibit the expected quantitative dose-response relationship; instead, it predicts a statistically significant reverse pattern where blocking tertiary amine metabolic liability sites increases (rather than decreases) predicted clearance.

Trajectory r36: The reverse dose-response relationship observed in the baseline HLM clearance model is driven by confounding physicochemical properties, specifically MolLogP, and is completely eliminated when molecular weight, MolLogP, and TPSA are included as explicit features alongside Morgan fingerprints.

Trajectory r55: The hypothesis is partially supported: in the confounded HLM fingerprint-only model, the top 20 most important fingerprint bits show low mean absolute TPSA correlation (0.1500) and no systematic relationship between TPSA correlation and clearance correlation ($r=0.122$, $p=0.608$), whereas in the mechan...

Trajectory r56: The hypothesis that TPSA is the key confounder is REJECTED; the actual confounding in HLM clearance predictions arises from MolWt and MolLogP, and only when these properties are properly controlled through multivariate matching does the combined-features model reduce systematic bias to non-significa...